KILNPULSE: MONITORING WHETHER INDIVIDUAL BRICK KILNS ARE RUNNING WITH SENTINEL-1, SENTINEL-2 AND LANDSAT

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ABSTRACT

Brick kilns are a major source of air pollution in Pakistan, and Punjab orders them shut during the winter smog season. Maps now show where kilns are, but not which ones are running. We present KilnPulse, which estimates for each mapped kiln, in every 12-day window, whether it is running, using only free data in Google Earth Engine. Every clue compares a kiln with its own surrounding fields: radar change in the kiln yard from Sentinel-1, which sees through smog and cloud; vegetation, brick-red and dark-surface contrasts from Sentinel-2; and kiln warmth from Landsat 8/9. A Random Forest combines the clues and a two-state hidden Markov model smooths each kiln's sequence. On 300 kilns in Kasur and Sheikhupura districts over two seasons, with **[1]** kiln-windows labelled from very-high-resolution imagery by two annotators (Cohen's κ = **[2]**), KilnPulse reached F1 **[2]** (ROC-AUC **[2]**) in cross-validation grouped by kiln, against **[3]** for a calendar baseline. In smog-season windows, radar alone reached ROC-AUC **[4]**. Code and labels are public.

Index Terms— Brick kilns, activity monitoring, Sentinel-1, Sentinel-2, Landsat, air pollution, hidden Markov model

1. INTRODUCTION

South Asia's brick belt holds tens of thousands of kilns: one benchmark maps 62,671 across the region [1], and an inventory of Pakistan's part of the Indo-Gangetic Plain (IGP) lists about 11,000 [2]. Brick kilns are a major source of air pollution in Pakistan, many operating without regulation [2], and kilns are known sites of bonded labour [3]. To cut winter smog, Pun-jab orders kilns to stop: in 2020 the provincial Environment Protection Agency told the Lahore High Court that all kilns would close from 7 November to 31 December [4], and in the 2025 smog season the province reported 3,178 kilns sealed and 485 demolished [5]. Checking such orders means visiting kilns one at a time.

Satellite studies have focused on where kilns are. Deep learning finds kilns at national scale [1, 2, 6], including their type [1, 2], and Patel et al. [7] combine detections with siting rules to flag non-compliant kilns. When kilns run is far less studied. Misra et al. [8] used Landsat NDVI histories to date when kilns were built or abandoned, noting that idle kilns grow grass, and Nazir et al. [9] flagged smog-season kiln activity from coarse-resolution thermal and trace-gas data. Patel et al. write that "identifying whether a kiln is operational at a given time remains challenging with RGB imagery" [7]. The problem is hardest in exactly the months when closures are ordered, because smog and fog hide the ground from optical sensors.

A running kiln, however, changes its surroundings in ways that free 10–30 m sensors can see. Workers mould and dry rows of green bricks, move stacks of fired bricks and pile coal in the yard; such changes alter the radar backscatter between Sentinel-1 passes, and C-band radar sees through cloud and haze [10]. Production also keeps vegetation off the kiln and yard and exposes red brick and dark coal that Sentinel-2 can measure [11], and firing may warm the kiln surface in Landsat thermal data. The short-wave infrared (SWIR) hot-spot indices used for volcanoes [12] look for very hot exposed surfaces; in a kiln the fire burns inside the brick setting under a layer of ash, seen from above only through small fuel holes [7], so we test whether such an index works at all.

Our contributions are: (i) KilnPulse, a per-kiln estimate of whether each kiln is running in every 12-day window, built from kiln-versus-background contrasts in free radar, optical and thermal data; (ii) a two-annotator label set for 300 kilns over two seasons; (iii) a test, with hypotheses fixed before labelling, of what each sensor adds, including in smog and in windows without a clear optical view, and of transfer to an unseen district and year; and (iv) district time series of the share of kilns running, compared with closure periods.

REFERENCES

[1] R. Mondal, J. Parab, H. Kubadia, S. Dubey, S. Junagade,
Z. B. Patel, and N. Batra, “SentinelKilnDB: A large-scale dataset and benchmark for OBB brick kiln detection in South Asia using satellite imagery,” in *Advances in Neural Information Processing Systems (NeurIPS)*, 2025.

[2] M. S. A. Hamdani, K. Zakir, N. Kushwaha, S. E. Fatima, and H. A. Sheikh, “Brick kiln dataset for Pakistan’s IGP region using AI,” *Scientific Data*, vol. 12, p. 830, 2025.

[3] D. S. Boyd, B. Jackson, J. Wardlaw, G. M. Foody,
S. Marsh, and K. Bales, “Slavery from space: Demon-strating the role for satellite remote sensing to inform evidence-based action related to UN SDG number 8,” *ISPRS Journal of Photogrammetry and Remote Sensing*, vol. 142, pp. 380–388, 2018.

[4] H. Hasan, “Smog control measures: Kilns to be closed for two months,” Zameen News, https://www.zameen.c om/news/smog-control-measures-kilns-to-be-closed-f or-two-months.html, Sep. 2020.

[5] Environment Protection Department, Government of the Punjab, “Smog mitigation,” https://epd.punjab.gov.pk/no de/5569, Dec. 2025.

[6] J. Lee, N. R. Brooks, F. Tajwar, M. Burke, S. Ermon,
D. B. Lobell, D. Biswas, and S. P. Luby, “Scalable deep learning to identify brick kilns and aid regulatory capac-ity,” *Proceedings of the National Academy of Sciences*, vol. 118, no. 17, p. e2018863118, 2021.

[7] Z. B. Patel, R. Mondal, S. Dubey, S. Jaiswal, S. Gut-tikunda, and N. Batra, “Space to Policy: Scalable brick kiln detection and automatic compliance monitoring with geospatial data,” arXiv:2412.04065, 2024.